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The effectiveness of educational interventions to develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students: a systematic review protocol

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The effectiveness of educational interventions to develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students: a systematic review protocol

Corresponding Author

Helena De Rezende. Kingston University and St George's University of London, Room FL 1015, Frank Lampl Building, Kingston Hill Campus, Kingston upon Thames, KT2 7LB, United Kingdom; Email: h.derezende@sgul.kingston.ac.uk

Co-Authors

Aline Mirema F. Vitorio, State Superintendence of the Ministry of Health in Goiás, Ministry of Health. Goiânia, Brazil.

Alexandre Souza Morais, Department of Health Sciences, Federal University of Espírito Santo. Vitória, Brazil.

Ana Claudia A. Garzin, School of Nursing, São Camilo University Center. São Paulo, Brazil.

Andressa Garcia Nicole, Department of Health Sciences, Federal University of Espírito Santo. Vitória, Brazil.

Ellen Regina Sevilla Quadrado, Quality and Safety in Nursing and Health Services Research Group, University of São Paulo. São Paulo, Brazil.

Daniela Campos de Andrade Lourenção, School of Nursing, University of São Paulo. São Paulo, Brazil.

Maristela Santini Martins, School of Nursing, University of São Paulo. São Paulo, Brazil.

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The effectiveness of educational interventions to develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students: a systematic review protocol

Abstract

Introduction: Despite global efforts taken in the last two decades to reduce adverse events, unsafe care is one of the leading causes of patient death and disability. Given the size of the nursing workforce in the healthcare system the inclusion of patient safety in the undergraduate nursing curriculum is necessary to enhance a safe culture in the daily work of their future careers. To this end, it is essential to apply effective teaching strategies to develop patient safety competencies. Thus, this review will aim to evaluate the effectiveness of educational interventions in developing patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students.

Methods and analysis: This protocol has been approved by the Joanna Briggs Institute Methodology for Systematic Reviews of Effectiveness and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols guidelines. Published studies will be searched across seven databases from 2011 to the present, along with a comprehensive search of grey literature. Two independent reviewers will conduct the search, extract the data, and assess the risk of bias for the included studies, using standardised critical appraisal instruments from the Joanna Briggs Institute. The quality of the evidence will be assessed using the Grading of Recommendations, Assessment Development and Evaluation (GRADE) methodology. Studies will be pooled in the meta-analysis. Alternatively, the findings will be presented in narrative form, including tables and figures, to aid in data presentation.

Ethics and dissemination: This study raises no ethical issues. The findings will be disseminated through presentations at professional conferences and publications in a peer-reviewed journal.

PROSPERO registration number: CRD 42021254965.

Keywords: Nursing Students, Education, Patient Safety

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Article Summary

Strengths and limitations of this study

- This study may potentially inform the literature gaps and help nursing educators make decisions regarding the best approach for teaching patient safety to nursing students.
- The review will adhere to the Joanna Briggs Institute Methodology for Systematic Reviews of Effectiveness and Preferred Reporting Items for Systematic Reviews, to ensure a rigorous and systematic approach to searching, screening, and reporting.
- The certainty of the evidence for all outcomes will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology.
- Limiting the population only to nursing students may not capture studies that have applied educational interventions combined with students from other health courses.
- There may be significant heterogeneity owing to the different educational interventions, duration, and frequency of the courses to teach patient safety.

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INTRODUCTION

Patient safety is a global and strategic priority in health systems. Regardless of all the efforts made in the last two decades to reduce and prevent errors, recent studies suggest that unsafe care is one of the leading causes of morbidity and mortality worldwide[1,2]. Although patient safety has a greater focus on hospital settings, it has been a systemic issue since 20%–25% of the general population has experienced harm in primary and outpatient care settings[3,4].

Several studies have reported a broader impact of adverse events. Hospitalisations in low and middle-income countries cause 134 million adverse events each year, contributing to more than 2.5 million deaths annually[5]. An analysis by the Organisation for Economic Cooperation and Development (OECD) has found that 15% of all hospital costs in its member states are due to patient harm caused by adverse events[6]. Worldwide the social cost of patient harm can be valued at US$ 1 trillion to 2 trillion a year[7].

To reduce the risks and the incidence of these avoidable incidents, making the error less likely and decreasing its impact, the World Health Organisation (WHO) considers the need for collaborative activities that create culture, behaviours, processes, procedures, technologies, and environments towards patient safety[7]. Thus, it is essential to develop interventions that can incorporate patient safety topics in the training of health professionals to help them cultivate competencies for safe care.

Considerable importance should be given to nursing students, given the size of the nursing workforce in the most varied health care settings as well as their key position in the provision and coordination of care, and participation in organisational, quality, and safety structures with optimisation of patient outcomes[8-10].

Owing to these critical roles, undergraduate nursing education must develop future nurses' knowledge, skills, behaviours, and attitudes that can help their adherence to patient safety principles[11] and improve the quality of healthcare systems[12,13].

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Various initiatives have been established to support the development of these core dimensions and inform faculty resources designed to teach patient safety. In 2005 the Quality and Safety Education for Nurses (QSEN) project, has proposed targets for quality and safety knowledge, skills, and attitudes to be developed for undergraduate nursing programmes[14,15]. In 2011, the WHO has published the Patient Safety Curriculum Guide: Multi-professional Edition which has provided a comprehensive curriculum designed to create a solid foundation of knowledge and skills to enable future healthcare professionals to present safe attitudes and behaviours in different healthcare settings[16]. In addition, it has addressed pedagogical principles and teaching strategies to contribute to implementing the subject and promoting meaningful student learning[16]. More recently, in 2017, the Patient Safety Competency Framework for Nursing Students was developed in Australia to provide key patient safety competencies statements and significant knowledge and skills to nursing courses[17].

Many nursing pre-licensure education programmes utilise these frameworks. Despite the increasing interest in the necessary knowledge, skills, attitudes, and behaviours surrounding patient safety, a considerable amount of literature has indicated that teaching patient safety is still inconsistent in nursing education[2, 18-20]. Furthermore, there is a lack of consensus on how patient safety contents should be effectively taught to pre-registration nursing students and what teaching methods would be employed[21-24].

In light of this evidence, it is imperative to investigate the effectiveness of educational interventions that might be useful in redesigning curricula to address nursing student learning needs concerning patient safety, promote this knowledge, and recommend implementation strategies.

A preliminary search of the Cochrane Database of Systematic Reviews, JBI Database of Systematic Reviews and Implementation Reports, PROSPERO, and MEDLINE, revealed the existence of two systematic reviews published on this topic[25, 26].

The rapid review published by Bianch et al.[25] has investigated the clinical learning environments that facilitate nursing students' development of patient safety

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competencies. However, it did not include teaching strategies in classroom environments that could enable these competencies; additionally, grey literature was not searched.

A recent review by Lee et al.[26] aimed to identify tested patient safety interventions that are helpful in teaching nursing students. However, this study has some methodological limitations. The authors did not publish a protocol, search for reference lists or grey literature, and only articles in English were included. In addition, they did not evaluate the quality of the body of evidence. Because the authors have only explored the core concepts of patient safety, only four articles investigating nursing students have been published. Hence, they further included studies of pre-licensure nursing students, in addition to students from other disciplines.

Given these limitations, we present a protocol for a systematic review that aims to evaluate the effectiveness of educational interventions delivered in both clinical and university settings that can develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students. To have a wider picture of the effects of the teaching methods, we will include studies that have explored the teaching of any patient safety content within the existing topic areas of the WHO Multi-professional Patient Safety Curriculum Guide[16]. We also aim to find available evidence through a comprehensive search, including published and unpublished studies in English, Portuguese, and Spanish with a systematic approach to screening and reporting. In addition, the certainty of evidence for all outcomes will be assessed using the Grading of Recommendations, Assessment, Development and Evaluation framework (GRADE)[27].

METHODS AND ANALYSIS

The protocol for this review was developed under the Joanna Briggs Institute Methodology for Systematic Reviews of Effectiveness[28] and Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P)[29].

The protocol was registered on the International Prospective Register of Systematic Reviews (PROSPERO) with the number CRD42021254965 on the first of June 2021. If

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amendments to this protocol are required, these will be recorded in PROSPERO with a description of the change and its rationale.

Review Question

What is the effectiveness of educational interventions in developing patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students?

Inclusion Criteria

The inclusion and exclusion criteria of the studies to be included in the review are detailed using the PICOS (P: Population; I: Indicator/Intervention; C: Comparator; O: Outcome(s); S: Study design) framework[30, 31] (Table 1).

Population

This review will include undergraduate nursing students of any age or gender undertaking a full or part-time programme of study. All years of study will be included in the review and, when applicable, all fields of nursing.

Interventions

This review will consider studies that evaluate any educational intervention aimed at teaching patient safety contents within the existing topic areas of the WHO Multi-professional Patient Safety Curriculum Guide[16]. This may employ the following teaching methods, but are not limited to, lectures, clinical placements, online activities, problem-based learning, simulations/skills laboratories, tutorials, workshops, group discussions and other strategies[16].

The intervention may take place in a tertiary education environment, clinical setting, or both. No limitations will be applied to the programme duration or intensity.

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Comparator

This review will include studies that compare educational interventions with alternative or different interventions or the absence of educational interventions.

Outcomes

This review will consider studies that reported at least one of the following outcomes: nursing students' knowledge, skills, attitudes, and behaviours related to patient safety[16, 17, 32] assessed by any validated instrument.

If outcomes are assessed using non-validated instruments, they will be synthesised narratively in the final report.

Study Design

This review will consider experimental and quasi-experimental study designs, including randomised controlled trials, non-randomised controlled trials, before-and-after studies, and interrupted time-series studies. Additionally, analytical observational studies, including prospective and retrospective cohort studies, case-control studies, and analytical cross-sectional studies, will be considered for inclusion.

Table 1 Description of inclusion and exclusion criteria.

Criterion	Definition	Rationale
Papers include sufficient empirical data.	It is not a review, commentary, letter, editorial and conference paper, or research report without the full text.	Primary studies with full data minimise the risk of bias and provide sufficient information for data extraction and quality assessment.
Participants include undergraduate nursing students.	Nursing students of any age or gender undertaking a full or part-time programme of study. All years of study will be included and, when applicable, all fields of nursing.	Undergraduate nursing students are the target population of this review.

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	It does not involve nursing associate students and mixed health students, even if it includes nursing students.	
Study involves an educational intervention.	Study reports an educational intervention offered to participants using any teaching methods[16]. It is not a study involving new strategies without an educational intervention.	Explicit educational interventions are the focus of this review.
Educational intervention includes patient safety as a core content.	Educational intervention includes any patient safety content within the topic areas of the WHO Multi-professional Patient Safety Curriculum Guide[16]: Topic 1: What is patient safety? Topic 2: Why applying human factors is important for patient safety. Topic 3: Understanding systems and the effect of complexity on patient care. Topic 4: Being an effective team player. Topic 5: Learning from errors to prevent harm. Topic 6: Understanding and managing clinical risk. Topic 7: Using quality improvement methods to improve care. Topic 8: Engaging with patients and carers. Topic 9: Infection prevention and control. Topic 10: Patient safety and invasive procedures. Topic 11: Improving medication safety. It is not a study that assesses the development of specific knowledge, skills, attitudes, and behaviours that does not have the primary focus on patient safety.	Educational interventions teaching patient safety contents within the WHO Multi-Professional Patient Safety Curriculum Guide are the focus of this review.
Study includes an evaluation of the educational intervention	Intervention is evaluated with regards to at least one of the following outcomes: knowledge, skills, attitudes, and behaviours related to patient safety[16, 17, 32]. It is not a purely descriptive study.	Allows comparative analysis of the effectiveness of interventions where possible.

Search Strategy

A three-step search strategy will be applied to locate published and unpublished studies[33].

An initial limited search of MEDLINE (PubMed) and CINAHL (EBSCOhost) was undertaken to identify articles on the topic. The text words contained in the titles and abstracts of relevant articles and the index terms used to describe the papers were used to develop a full search strategy for MEDLINE (PubMed) (see Online Supplemental

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Material Appendix 1 – Search Strategy Example). A second search, including all identified keywords and index terms, will be adapted for each included information source. Third, the reference lists of all studies selected for critical appraisal will be screened for additional studies. Studies published in English, Spanish, and Portuguese will be included.

Studies published from 2011 to the present will be included. The rationale for this data was when the National Patient Safety Curriculum Guide: a multi-professional edition (WHO, 2011) was published. The guide provides the most relevant evidence on the development and implementation of global patient safety education initiatives carried out by undergraduate healthcare students.

The databases to be searched include MEDLINE (PubMed), CINAHL (EBSCOhost), Scopus (Elsevier), Education Research Complete (EBSCOhost), The Cochrane Central Register of Controlled Trials (CENTRAL), Latin American and Caribbean Health Sciences Literature (LILACS), and Medes (Spain).

Sources of unpublished studies and grey literature to be searched will include ClinicalTrials.gov, Google Scholar, OpenGrey, ProQuest Dissertations and Theses, Coordination for the Improvement of Higher Education Personnel - Brazil (CAPES thesis and dissertations), The Virginia Henderson Global e-Repository, Mednar, and Thesis Canada.

Study Selection

Following the search, all identified citations will be collated and uploaded to EndNote online (Clarivate Analytics, PA, USA), and duplicates will be removed.

Following a pilot test, titles and abstracts will be screened by two independent reviewers for assessment of the inclusion criteria for the review. Potentially relevant studies will be retrieved in full, and their citation details will be imported into the JBI System for the Unified Management, Assessment, and Review of Information (JBI SUMARI; JBI, Adelaide, Australia)[34].

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The full text of the selected citations will be assessed in detail against the inclusion criteria by two independent reviewers. The reasons for excluding full-text studies that did not meet the inclusion criteria will be recorded and reported in the systematic review. Any disagreements between the reviewers at each stage of the study selection process will be resolved through discussion or by a third reviewer.

The search results, study selection, and inclusion process will be reported in full in the final systematic review and presented in preferred reporting items for systematic reviews and meta-analyses (PRISMA) flow diagram[35].

Assessment of Methodological Quality

Two independent reviewers will critically appraise eligible studies at the study level for methodological quality in the review using standardised critical appraisal instruments from JBI for experimental, quasi-experimental, and observational studies (see Online Supplemental Material Appendix 2 – Critical Appraisal Tools)[28]. Authors of papers will be contacted to request missing or additional data for clarification, where required.

Any disagreements that arise will be resolved through discussion or with a third reviewer.

Regardless of methodological quality, all studies will undergo data extraction and synthesis (where possible). A table accompanying a narrative will report the results of critical appraisal.

Data Extraction

Data will be extracted from studies included in the review by two independent reviewers using an adapted version of the JBI data extraction tool[28], including specific information related to the characteristics of the intervention (see Online Supplemental Material Appendix 3 –Extraction Tool). The data extract will also include specific details about the study author(s), country, characteristics of higher education institution, setting of the intervention, characteristics of participants (including year and field of study), study design, description of the intervention (including the type of educational method and

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duration of the intervention), outcomes of significance to the review question assessed (knowledge, skills, attitudes, and behaviours), name and type of instrument used to evaluate the intervention, follow-up time, main results, limitations, and additional data when required.

Data Synthesis

Literature has suggested that it is challenging to synthesise data from educational interventions owing to heterogeneities in interventions and studies methodologies[36, 37]. Therefore, data will be initially assessed for similarity or extent of variation in outcome measures, measurement scales, and type of interventions before determining whether it is appropriate to enter into a meta-analysis. If it is considered suitable, data will be entered into a meta-analysis using JBI SUMARI.

Effect sizes will be expressed as either odds ratios (for dichotomous data) or weighted (or standardised) final post-intervention mean differences (for continuous data), and their 95% confidence intervals will be calculated for analysis. If meta-analysis can be conducted, heterogeneity will be assessed statistically using the standard χ^2 and I^2 tests. Statistical analyses will be performed using a random effects model[38] to allow generalisation.

To investigate potential sources of heterogeneity, subgroup analyses will be performed based on the year of the study, the field of nursing if applicable, intervention delivery (teaching method), the content of the intervention (patient safety topics), and the time frame of the intervention. Sensitivity analyses will be conducted to test decisions made regarding the effectiveness of interventions. If there is significant variation in the data and statistical pooling is not possible, the findings will be presented in narrative form, including tables and figures to aid in data presentation[39].

A funnel plot will be generated using RevMan 5.3 (Copenhagen: The Nordic Cochrane Centre, Cochrane) to assess publication bias if ten or more studies are included in the meta-analysis. Statistical tests for funnel plot asymmetry (Egger test) will be performed where appropriate.

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Assessing Certainty of Findings

The GRADE approach for grading the certainty of evidence will be followed[27], and a summary of findings will be created using GRADEpro GDT V5 (McMaster University, ON, Canada).

The Summary of Findings will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence based on the risk of bias, directness, heterogeneity, precision, and risk of publication bias of the review results. The outcomes reported in the Summary of Findings will be knowledge, skills, attitudes, and behaviours of undergraduate nursing students related to patient safety.

PATIENT AND PUBLIC INVOLVEMENT

There will be no patient or public involvement in the conduct or dissemination of the results of this study.

ETHICS AND DISSEMINATION

This review does not require ethical approval because the primary population data will not be collected. This protocol complies with PRISMA guidelines. The findings will be disseminated at professional conference presentations and publications in pre-print and peer-reviewed open access journals.

COMPETING INTERESTS

The authors declare no conflict of interest, financial or otherwise.

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AUTHORS' CONTRIBUTIONS

HDR conceptualised the study with inputs from AG and AN. HDR designed and drafted the review protocol. AM, AV, AG, AN, EQ, and DL contributed to the protocol development. MS provided critical revision of the drafts for important intellectual content. All authors provided input to the drafts of the manuscript and approved the final version of this manuscript.

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REFERENCES

1. World Health Organization (WHO). Ten facts on patient safety. 2019.Available https://www.who.int/news-room/facts-in-pictures/detail/patient-safety[accessed July 2021].

2. Wu AW, Busch IM. Patient safety: a new basic science for professional education. *GMS J Med Educ* 2019;36(2):Doc21.doi:10.3205/zma001229.

3.Bates DW, Singh H. Two Decades Since To Err Is Human: An Assessment Of Progress And Emerging Priorities In Patient Safety. *Health Aff* 2018;37(11):1736-43. doi:10.1377/hlthaff.2018.0738.

4. Kuriakose R, Aggarwal A, Sohi RK, et al. Patient safety in primary and outpatient health care. *J Family Med Prim Care* 2020;9:7-11.doi:10.4103/jfmpc.jfmpc_837_19.

5. National Academies of Sciences, Engineering, and Medicine- EAM. Crossing the global quality chasm: Improving health care worldwide. Washington, DC: The National Academies Press 2018.doi:10.17226/2515.

6. Slawomirski L, Auraaen A, Klazinga N. The economics of patient safety: Strengthening a value-based approach to reducing patient harm at national level. *OECD Health Working Papers* 2017;96.doi: 10.1787/5a9858cd-en.

7. World Health Organization (WHO). Global Patient Safety Action Plan 2021–2030: Towards eliminating avoidable harm in health care. Geneva, Switzerland: World Health Organization 2021.Availablehttps://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan [accessed Jan 2021].

8. Aiken LH, Sloane DM, Bruyneel L, et al. Nurse staffing and education and hospital mortality in nine European countries: a retrospective observational study. *Lancet* 2014;383(9931):1824-30. doi:10.1016/S0140-6736(13)62631-8 .

9. Oldland E, Botti M, Hutchinson AM, et al. A framework of nurses' responsibilities for quality healthcare — Exploration of content validity. *Collegian* 2020;27(2):150-63. doi:10.1016/j.colegn.2019.07.007[published Online First: 20 November 2019].

10. World Health Organization (WHO). Nursing and midwifery. 2020.https://www.who.int/news-room/fact-sheets/detail/nursing-and-midwifery[accessed May 2021].

11. Vaismoradi M, Tella S, Logan PA, et al. Nurses' Adherence to Patient Safety Principles: A Systematic Review. *Int J Environ Res Public Health* 2020;17(6):2028.doi: 10.3390/ijerph17062028.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12. Mansour MJ, Al Shadafan SF, Abu- Sneineh FT, AlAmer MM. Integrating Patient Safety Education in the Undergraduate Nursing Curriculum: A discussion Paper. *Open Nurs J* 2018;12:125-32.doi:10.2174/1874434601812010125.

13. Dimitriadou M, Merkouris A, Charalambous A, et al. The knowledge about patient safety among undergraduate nurse students in Cyprus and Greece: a comparative study. *BMC Nurs* 2021;20(110).doi:10.1186/s12912-021-00610-6.

14. QSEN Institute. https://qsen.org/ (accessed July 2021).

15. Cronenwett L, Sherwood G, Barnsteine J, et al. Quality and safety education for nurses. *Nurs Outlook* 2007;55(3):122-31.doi:10.1016/j.outlook.2007.02.006.

16. World Health Organization (WHO). WHO Patient safety curriculum guide: Multi-professional edition. Geneva, Switzerland: World Health Organization 2011.https://apps.who.int/iris/bitstream/handle/10665/44641/9789241501958_eng.pdf (accessed Jul 2021).

17. Levett-Jones T, Dwyer T, Reid-Searl K, et al. Patient safety competency framework (PSCF) for nursing students. 2017.https://www.cqu.edu.au/__data/assets/pdf_file/0026/65780/PatientSafetyCompetencyFrameworkFINAL.pdf(accessed Jul 2021).

18. Murray M, Sundin D, Cope V. New graduate registered nurses' knowledge of patient safety and practice: A literature review. *J Clin Nurs* 2018;27:31-47.doi:10.1111/jocn.13785.

19. Garzin ACA, Melleiro MM. Safety in the training of health professionals. *Cienc Cuid Saude* 2019;18(4):e45780.doi:10.4025/cienccuidsaude.v18i4.45780.

20. Rebeschi LM. Perceived patient safety competence of baccalaureate nursing students: A descriptive comparative study. *SAGE Open Nurs* 2020;6:2377960820930134. doi:10.1177/2377960820930134

21. Tella S, Liukka M, Jamookeeah D, et al. What do nursing students learn about patient safety? An integrative literature review. *J Nurs Educ* 2014;53(1):7-13.doi:10.3928/01484834-20131209-04.

22. Usher k, Woods C, Conway J, et al. Patient safety contente and delivery in pre-registration nursing curricula: A national cross-sectional survey study. *Nurse Educ Today* 2018;66:82-89.doi:10.1016/j.nedt.2018.04.013.

23. Kirwan M, Riklikiene O, Gotlib J, et al. Regulation and current status of patient safety content in pre-registration nurse education in 27 countries: Findings from the rationing - missed nursing care (RANCARE) COST action project. *Nurse Educ Pract* 2019;37:132-140.doi:10.1016/j.nepr.2019.04.013.

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24. VanGraafeiland B, Sloand E, Silbert-Flagg J, et al. Academic-clinical service partnerships are innovative strategies to advance patient safety competence and leadership in prelicensure nursing students. *Nurs Outlook* 2019;67(1):49-53.doi:10.1016/j.outlook.2018.08.003 .

25. Bianchi M, Bressan V, Cadorin L, et al. Patient safety competencies in undergraduate nursing students: a rapid evidence assessment. *J Adv Nurs* 2016;72(12):2966-2979.doi:10.1111/jan.13033.

26. Lee SE, Morse BL, Kim NW. Patient safety educational interventions: A systematic review with recommendations for nurse educators. *Nurs Open* 2021;00:1–13.doi:10.1002/nop2.955.

27. Guyatt G, Oxman AD, Akl EA, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol* 2011;64(4):383–94.doi:10.1016/j.jclinepi.2010.04.026.

28. Tufanaru C, Munn Z, Aromataris E, et al. Chapter 3: Systematic reviews of effectiveness. In: Aromataris E, Munn Z, eds. JBI Manual for Evidence Synthesis: JBI 2020.https://synthesismanual.jbi.global/ (accessed Jun 2021).

29. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71.doi:10.1136/bmj.n71.

30. McKenzie JE, Brennan SE, Ryan RE, et al. Chapter 3: Defining the criteria for including studies and how they will be grouped for the synthesis. In: Higgins JPT, Thomas J, Chandler J, et al, eds. Cochrane Handbook for Systematic Reviews of Interventions version 6.2 (updated February 2021): Cochrane 2021.www.training.cochrane.org/handbook (accessed Jul 2021).

31. Stern C, Jordan Z, McArthur A. Developing the review question and inclusion criteria. *Am J Nurs* 2014;114(4):53-6.doi:10.1097/01.NAJ.0000445689.67800.86.

32. Yu A. Fontana G. Darzi A. Evaluation and training interventions for patient safety: A report by the centre for health policy at Imperial College London, an academic partner to Health Education England and the commission on education and training for patient safety 2016. https://www.hee.nhs.uk/sites/default/files/documents/Evaluation%20of%20Education%20and%20Training%20Interventions%20for%20Patient%20Safety%20-%20Imperial%20College%20London.pdf(accessed Jul 2021).

33. Aromataris E, Munn Z (Editors). JBI Manual for Evidence Synthesis. JBI, 2020.https://synthesismanual.jbi.global (accessed Jun 2021).

34. Munn Z, Aromataris E, Tufanaru C, et al. The development of software to support multiple systematic review types: the Joanna Briggs Institute System for the Unified

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Management, Assessment and Review of Information (JBI SUMARI). *Int J Evid Based Healthc* 2019;17(1):36-43.doi:10.1097/xeb.0000000000000152.

35. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71.doi:10.1136/bmj.n71.

36. Reed D, Price EG, Windish DM, et al. Challenges in systematic reviews of educational intervention studies. *Ann Intern Med* 2005;142(12Pt2):1080-9.doi:10.7326/0003-4819-142-12_Part_2-200506211-00008.

37. Slavin RE. Perspectives on evidence-based research in education—What works? Issues in synthesizing educational program evaluations. *Educ Res* 2008;37(1):5-14.doi:10.3102/0013189X08314117 [published Online First: 21 June 2005].

38. Tufanaru C, Munn Z, Stephenson M, et al. Fixed or random-effects meta-analysis? Common methodological issues in systematic reviews of effectiveness. *Int J Evid Based Healthc* 2015;13(3):196–207.doi:10.1097/xeb.0000000000000065.

39. Campbell M, McKenzie JE, Sowden A, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. BMJ. 2020; 368: l6890.doi:10.1136/bmj.l6890.

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Appendix 1

Search Strategy Example: MEDLINE (Pubmed)

#1"students, nursing"[MeSH Terms] OR "nursing students"[Title/Abstract] OR "pupil nurses"[Title/Abstract] OR "undergraduate nurses"[Title/Abstract] OR "baccalaureate nurses"[Title/Abstract] OR "student nurses"[Title/Abstract] OR "undergraduate student nurses"[Title/Abstract] OR ("pre-licensure"[All Fields] AND "nurses"[Title/Abstract])
Total 32.627

#2 "educational status"[MeSH Terms] OR "education"[MeSH Terms] OR "education"[Title/Abstract] OR ("educational"[All Fields] AND "activities"[Title/Abstract]) OR "training program*"[Title/Abstract] OR "learning"[Title/Abstract] OR "teaching"[Title/Abstract] OR "teaching strateg*"[Title/Abstract] OR "teaching methods"[Title/Abstract]
Total 1.503.612

#3 "patient safety"[MeSH Terms] OR "patient safety"[Title/Abstract] OR "patient safety competency"[Title/Abstract] OR "error"[Title/Abstract] OR "incident"[Title/Abstract]
Total 342.519

#1 AND #2 AND #3
Total 691

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Appendix 2

Critical Appraisal Tools

JBI CRITICAL APPRAISAL CHECKLIST FOR RANDOMIZED CONTROLLED TRIALS

Reviewer_____

Date_____

Author_____Year_____ Record Number_____

		Yes	No	Unclear	NA
1.	Was true randomization used for assignment of participants to treatment groups?	☐	☐	☐	☐
2.	Was allocation to treatment groups concealed?	☐	☐	☐	☐
3.	Were treatment groups similar at the baseline?	☐	☐	☐	☐
4.	Were participants blind to treatment assignment?	☐	☐	☐	☐
5.	Were those delivering treatment blind to treatment assignment?	☐	☐	☐	☐
6.	Were outcomes assessors blind to treatment assignment?	☐	☐	☐	☐
7.	Were treatment groups treated identically other than the intervention of interest?	☐	☐	☐	☐
8.	Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	☐	☐	☐	☐
9.	Were participants analyzed in the groups to which they were randomized?	☐	☐	☐	☐
10.	Were outcomes measured in the same way for treatment groups?	☐	☐	☐	☐
11.	Were outcomes measured in a reliable way?	☐	☐	☐	☐
12.	Was appropriate statistical analysis used?	☐	☐	☐	☐
13.	Was the trial design appropriate, and any deviations from the standard RCT design (individual randomization, parallel groups) accounted for in the conduct and analysis of the trial?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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JBI CRITICAL APPRAISAL CHECKLIST FOR QUASI-EXPERIMENTAL STUDIES

Reviewer_____
Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	□	□	□	□
2. Were the participants included in any comparisons similar?	□	□	□	□
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	□	□	□	□
4. Was there a control group?	□	□	□	□
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	□	□	□	□
6. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	□	□	□	□
7. Were the outcomes of participants included in any comparisons measured in the same way?	□	□	□	□
8. Were outcomes measured in a reliable way?	□	□	□	□
9. Was appropriate statistical analysis used?	□	□	□	□

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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JBI CRITICAL APPRAISAL CHECKLIST FOR CASE SERIES

Reviewer_____
Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
• Were there clear criteria for inclusion in the case series?	□	□	□	□
• Was the condition measured in a standard, reliable way for all participants included in the case series?	□	□	□	□
• Were valid methods used for identification of the condition for all participants included in the case series?	□	□	□	□
• Did the case series have consecutive inclusion of participants?	□	□	□	□
• Did the case series have complete inclusion of participants?	□	□	□	□
• Was there clear reporting of the demographics of the participants in the study?	□	□	□	□
• Was there clear reporting of clinical information of the participants?	□	□	□	□
• Were the outcomes or follow up results of cases clearly reported?	□	□	□	□
• Was there clear reporting of the presenting site(s)/clinic(s) demographic information?	□	□	□	□
• Was statistical analysis appropriate?	□	□	□	□

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

JBI CRITICAL APPRAISAL CHECKLIST FOR COHORT STUDIES

For peer review only

Reviewer_____

Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the two groups similar and recruited from the same population?	☐	☐	☐	☐
2. Were the exposures measured similarly to assign people to both exposed and unexposed groups?	☐	☐	☐	☐
3. Was the exposure measured in a valid and reliable way?	☐	☐	☐	☐
4. Were confounding factors identified?	☐	☐	☐	☐
5. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	☐	☐	☐	☐
7. Were the outcomes measured in a valid and reliable way?	☐	☐	☐	☐
8. Was the follow up time reported and sufficient to be long enough for outcomes to occur?	☐	☐	☐	☐
9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	☐	☐	☐	☐
10. Were strategies to address incomplete follow up utilized?	☐	☐	☐	☐
11. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

For peer review only

JBI CRITICAL APPRAISAL CHECKLIST FOR CASE CONTROL STUDIES

Reviewer_____

Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the groups comparable other than the presence of disease in cases or the absence of disease in controls?	□	□	□	□
2. Were cases and controls matched appropriately?	□	□	□	□
3. Were the same criteria used for identification of cases and controls?	□	□	□	□
4. Was exposure measured in a standard, valid and reliable way?	□	□	□	□
5. Was exposure measured in the same way for cases and controls?	□	□	□	□
6. Were confounding factors identified?	□	□	□	□
7. Were strategies to deal with confounding factors stated?	□	□	□	□
8. Were outcomes assessed in a standard, valid and reliable way for cases and controls?	□	□	□	□
9. Was the exposure period of interest long enough to be meaningful?	□	□	□	□
10. Was appropriate statistical analysis used?	□	□	□	□

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

_____

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JBI CRITICAL APPRAISAL CHECKLIST FOR ANALYTICAL CROSS SECTIONAL STUDIES

Reviewer_____
Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the criteria for inclusion in the sample clearly defined?	☐	☐	☐	☐
2. Were the study subjects and the setting described in detail?	☐	☐	☐	☐
3. Was the exposure measured in a valid and reliable way?	☐	☐	☐	☐
4. Were objective, standard criteria used for measurement of the condition?	☐	☐	☐	☐
5. Were confounding factors identified?	☐	☐	☐	☐
6. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
7. Were the outcomes measured in a valid and reliable way?	☐	☐	☐	☐
8. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____
_____
_____

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Appendix 3
Extraction Tool

Extraction Tool
Study refence (Author/year) and Study ID
Country: **Type of Higher Education Institution:** Public ( ) Private ( ) Other ( )
Setting of the intervention delivery: University ( ) Clinical Setting ( ) Both Combined ( )
Description of the intervention: Type of educational method (s): Duration of the intervention:
Group description: Intervention: Control:
Sample Size: Intervention: Control:
Outcomes of interest assessed: Knowledge ( ) Skills ( ) Attitudes ( ) Behaviours ( )
Name and description of instrument for outcome assessment:
Follow-up time:
Main Results:
Limitations:
Comments:

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Pages 7-8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Pages 9-10
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Online Supplemental Material Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 10
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 11
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Pages 11-12
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Pages 11-12
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 11
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 12
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 12
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 12
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 12
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 12
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 12
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 12
Reporting bias	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 12

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 13
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	N/A
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	N/A
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	N/A
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	N/A
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	N/A
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	N/A
	23b	Discuss any limitations of the evidence included in the review.	N/A
	23c	Discuss any limitations of the review processes used.	N/A
	23d	Discuss implications of the results for practice, policy, and future research.	N/A
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	N/A
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 7
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 13
Competing interests	26	Declare any competing interests of review authors.	Page 14
Availability of data, code and	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Online Supplemental Material

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
other materials			Appendices 2 and 3

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71
For more information, visit: http://www.prisma-statement.org/

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BMJ Open

The effectiveness of educational interventions to develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students: a systematic review protocol

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The effectiveness of educational interventions to develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students: a systematic review protocol

Corresponding Author

Helena De Rezende. Kingston University and St George's University of London, Room FL 1015, Frank Lampl Building, Kingston Hill Campus, Kingston upon Thames, KT2 7LB, United Kingdom; Email: h.derezende@sgul.kingston.ac.uk

Co-Authors

Aline Mirema F. Vitorio, State Superintendence of the Ministry of Health in Goiás, Ministry of Health. Goiânia, Brazil.

Alexandre Souza Morais, Department of Health Sciences, Federal University of Espírito Santo. Vitória, Brazil.

Ana Claudia A. Garzin, School of Nursing, São Camilo University Center. São Paulo, Brazil.

Andressa Garcia Nicole, Department of Health Sciences, Federal University of Espírito Santo. Vitória, Brazil.

Ellen Regina Sevilla Quadrado, Quality and Safety in Nursing and Health Services Research Group, University of São Paulo. São Paulo, Brazil.

Daniela Campos de Andrade Lourenção, School of Nursing, University of São Paulo. São Paulo, Brazil.

Maristela Santini Martins, School of Nursing, University of São Paulo. São Paulo, Brazil.

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The effectiveness of educational interventions to develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students: a systematic review protocol

Abstract

Introduction: Patient safety is a healthcare discipline that aims to prevent and reduce patient harm, risks and errors during the provision of healthcare. Given the size of the nursing workforce in the healthcare system the inclusion of patient safety in the undergraduate nursing curriculum is necessary to enhance a safe culture in the daily work of their future careers. To this end, it is essential to apply effective teaching strategies to develop patient safety competencies. This review will aim to evaluate the effectiveness of educational interventions in developing patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students within the existing topic areas of the World Health Organisation Multi-professional Patient Safety Curriculum Guide.

Methods and analysis: The databases Medline, CINAHL, Scopus, Education Research Complete, The Cochrane Central Register of Controlled Trials, LILACS, Medes and Grey literature such as ClinicalTrials.gov, Google Scholar, DART-Europe, ProQuest Dissertations, CAPES thesis and dissertations, The Virginia Henderson Global e-Repository, Mednar, and Thesis Canada will be searched from July 2011 until January 2022. Two independent reviewers will conduct the search, extract the data, and assess the risk of bias for the included studies, using standardised critical appraisal instruments from the Joanna Briggs Institute. The quality of the evidence will be assessed using the Grading of Recommendations, Assessment Development and Evaluation (GRADE) methodology. Studies will be pooled in the meta-analysis. Alternatively, the findings will be presented in narrative form, including tables and figures, to aid in data presentation.

Ethics and dissemination: This study raises no ethical issues. The findings will be disseminated through presentations at professional conferences and publications in a peer-reviewed journal.

PROSPERO registration number: CRD 42021254965.

Keywords: Nursing Students, Education, Patient Safety

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Article Summary

Strengths and limitations of this study

- The review will adhere to the Joanna Briggs Institute Methodology for Systematic Reviews of Effectiveness and Preferred Reporting Items for Systematic Reviews, to ensure a rigorous and systematic approach to searching, screening, and reporting.
- Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology will be used to assess the quality of evidence.
- Students from other health disciplines, such as medicine and allied health professions will not be included in this review; this may affect the results.
- Due to heterogeneity of educational interventions, there may be a limitation to perform quantitative synthesis.

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INTRODUCTION

Patient safety is a global and strategic priority in health systems and is defined *as "*a framework of organised activities that create cultures, processes, procedures, behaviours, technologies and environments in the health care that consistently and sustainably lower risks and reduce the occurrence of avoidable harm"[1 p.1].

Regardless of all the efforts made in the last two decades to reduce and prevent errors, recent studies suggest that unsafe care is one of the leading causes of morbidity and mortality worldwide[2,3]. Although patient safety has a greater focus on hospital settings, it has been a systemic issue since 20%–25% of the general population has experienced harm in primary and outpatient care settings[4,5].

Several studies have reported a broader impact of adverse events. Hospitalisations in low and middle-income countries cause 134 million adverse events each year, contributing to more than 2.5 million deaths annually[6]. An analysis by the Organisation for Economic Cooperation and Development (OECD) has found that 15% of all hospital costs in its member states are due to patient harm caused by adverse events[7]. Worldwide the social cost of patient harm can be valued at US$ 1 trillion to 2 trillion a year[1].

To reduce the risks and the incidence of these avoidable incidents, making the error less likely and decreasing its impact, the World Health Organisation (WHO) considers the need for collaborative activities that create culture, behaviours, processes, procedures, technologies, and environments towards patient safety[1]. Thus, it is essential to develop interventions that can incorporate patient safety topics in the training of health professionals to help them cultivate competencies for safe care.

Considerable importance should be given to nursing students, given the size of the nursing workforce in the most varied health care settings as well as their key position in the provision and coordination of care, and participation in organisational, quality, and safety structures with optimisation of patient outcomes[8-10].

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Owing to these critical roles, undergraduate nursing education must develop future nurses' knowledge, skills, behaviours, and attitudes that can help their adherence to patient safety principles[11] and improve the quality of healthcare systems[12,13].

Various initiatives have been established to support the development of these core dimensions and inform faculty resources designed to teach patient safety. In 2005 the Quality and Safety Education for Nurses (QSEN) project, proposed targets for quality and safety knowledge, skills, and attitudes to be developed for undergraduate nursing programmes[14,15]. In 2011, the WHO published the Patient Safety Curriculum Guide: Multi-professional Edition which provided a comprehensive curriculum designed to create a solid foundation of knowledge and skills to enable future healthcare professionals to present safe attitudes and behaviours in different healthcare settings[16]. In addition, it addressed pedagogical principles and teaching strategies to contribute to implementing the subject and promoting meaningful student learning[16]. More recently, in 2017, the Patient Safety Competency Framework for Nursing Students was developed in Australia to provide key patient safety competencies statements and significant knowledge and skills to nursing courses[17].

Many nursing pre-licensure education programmes utilise these frameworks. Despite the increasing interest in the necessary knowledge, skills, attitudes, and behaviours surrounding patient safety, a considerable amount of literature has indicated that teaching patient safety is still inconsistent in nursing education[2, 18-20]. Furthermore, there is a lack of consensus on how patient safety contents should be effectively taught to pre-registration nursing students and what teaching methods would be employed[21-24].

A preliminary search of the Cochrane Database of Systematic Reviews, JBI Database of Systematic Reviews and Implementation Reports, PROSPERO, and MEDLINE, revealed the existence of two systematic reviews published on this topic[25, 26].

The rapid review published by Bianch et al.[25] has investigated the clinical learning environments that facilitate nursing students' development of patient safety

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competencies. However, it did not include teaching strategies in classroom environments that could enable these competencies; additionally, grey literature was not searched.

A recent review by Lee et al.[26] aimed to identify tested patient safety interventions that are helpful in teaching nursing students. However, this study has some methodological limitations. The authors did not publish a protocol, search for reference lists or grey literature, and only articles in English were included. In addition, they did not evaluate the quality of the body of evidence. Because the authors had only explored the core concepts of patient safety, only four articles investigating nursing students have been published. Hence, they further included studies of pre-licensure nursing students, in addition to students from other disciplines.

Given these limitations, we present a protocol for a systematic review that aims to evaluate the effectiveness of educational interventions delivered in both clinical and university settings that can develop patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students. To have a wider picture of the effects of the teaching methods, we will include studies that have explored the teaching of any patient safety content within the existing topic areas of the WHO Multi-professional Patient Safety Curriculum Guide[16]. The guide provides the most relevant evidence on the development and implementation of global patient safety education initiatives carried out by undergraduate healthcare students.

METHODS AND ANALYSIS

The protocol for this review was developed under the Joanna Briggs Institute Methodology for Systematic Reviews of Effectiveness[27] and Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P)[28].

The protocol was registered on the International Prospective Register of Systematic Reviews (PROSPERO) with the number CRD42021254965 on the first of June 2021 and the planned date of the systematic review is June 2022. If amendments to this protocol

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are required, these will be recorded in PROSPERO with a description of the change and its rationale.

Review Question

How effective are educational interventions at developing patient safety knowledge, skills, behaviours, and attitudes in undergraduate nursing students?

Inclusion Criteria

The inclusion and exclusion criteria of the studies to be included in the review are detailed using the PICOS (P: Population; I: Indicator/Intervention; C: Comparator; O: Outcome(s); S: Study design) framework[29, 30] (Table 1).

Population

This review will include undergraduate nursing students of any age or gender undertaking a full or part-time programme of study. All years of study will be included in the review and, when applicable, all fields of nursing.

Interventions

This review will consider studies that evaluate any educational intervention aimed at teaching patient safety contents within the existing topic areas of the WHO Multi-professional Patient Safety Curriculum Guide[16]. This may employ the following teaching methods, but are not limited to, lectures, clinical placements, online activities, problem-based learning, simulations/skills laboratories, tutorials, workshops, group discussions and other strategies[16].

The intervention may take place in a tertiary education environment, clinical setting, or both. No limitations will be applied to the programme duration or intensity.

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Comparator

This review will include studies that compare educational interventions with alternative or different interventions or the absence of educational interventions.

Outcomes

This review will consider studies that described and evaluated at least one of the following outcomes: nursing students' knowledge, skills, attitudes, and behaviours related to patient safety[16, 17, 31].

Study Design

This review will consider experimental and quasi-experimental study designs, including randomised controlled trials, non-randomised controlled trials, before-and-after studies, and interrupted time-series studies. Additionally, analytical observational studies, including prospective and retrospective cohort studies, case-control studies, and analytical cross-sectional studies, will be considered for inclusion.

Table 1 Description of inclusion and exclusion criteria.

Criterion	Definition	Rationale
Papers include sufficient empirical data.	It is not a review, commentary, letter, editorial and conference paper, or research report without the full text.	Primary studies with full data minimise the risk of bias and provide sufficient information for data extraction and quality assessment.
Participants include undergraduate nursing students.	Nursing students of any age or gender undertaking a full or part-time programme of study. All years of study will be included and, when applicable, all fields of nursing. It does not involve nursing associate students and mixed health students, even if it includes nursing students.	Undergraduate nursing students are the target population of this review.

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Study involves an educational intervention.	Study reports an educational intervention offered to participants using any teaching methods[16]. It is not a study involving new strategies without an educational intervention.	Explicit educational interventions are the focus of this review.
Educational intervention includes patient safety as a core content.	Educational intervention includes any patient safety content within the topic areas of the WHO Multi-professional Patient Safety Curriculum Guide[16]: Topic 1: What is patient safety? Topic 2: Why applying human factors is important for patient safety. Topic 3: Understanding systems and the effect of complexity on patient care. Topic 4: Being an effective team player. Topic 5: Learning from errors to prevent harm. Topic 6: Understanding and managing clinical risk. Topic 7: Using quality improvement methods to improve care. Topic 8: Engaging with patients and carers. Topic 9: Infection prevention and control. Topic 10: Patient safety and invasive procedures. Topic 11: Improving medication safety. It is not a study that assesses the development of specific knowledge, skills, attitudes, and behaviours that does not have the primary focus on patient safety.	Educational interventions teaching patient safety contents within the WHO Multi-Professional Patient Safety Curriculum Guide are the focus of this review.
Study includes an evaluation of the educational intervention	Intervention is evaluated with regards to at least one of the following outcomes: knowledge, skills, attitudes, and behaviours related to patient safety[16, 17, 31]. It is not a purely descriptive study.	Allows comparative analysis of the effectiveness of interventions where possible.

Search Strategy

A three-step search strategy will be applied to locate published and unpublished studies[32].

An initial limited search of MEDLINE (PubMed) and CINAHL (EBSCOhost) was undertaken to identify articles on the topic. The text words contained in the titles and abstracts of relevant articles and the index terms used to describe the papers were used to develop a full search strategy for MEDLINE (PubMed) (see Online Supplemental Material Appendix 1 – Search Strategies). A second search, including all identified keywords and index terms, will be adapted for each included information source. Third,

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the reference lists of all studies selected for critical appraisal will be screened for additional studies. Studies published in English, Spanish, and Portuguese will be included because they are the languages of the systematic review team. If there is an English translation available, studies in other languages will be included.

Studies published from July 2011 to January 2022 will be included. The rationale for the date of July 2011 was when the WHO National Patient Safety Curriculum Guide: a multi-professional edition[16] was published.

The databases to be searched include MEDLINE (PubMed), CINAHL (EBSCOhost), Scopus (Elsevier), Education Research Complete (EBSCOhost), The Cochrane Central Register of Controlled Trials (CENTRAL), Latin American and Caribbean Health Sciences Literature (LILACS), and Medes (Spain).

Sources of unpublished studies and grey literature to be searched will include ClinicalTrials.gov, Google Scholar, DART-Europe, ProQuest Dissertations and Theses, Coordination for the Improvement of Higher Education Personnel - Brazil (CAPES thesis and dissertations), The Virginia Henderson Global e-Repository, Mednar, and Thesis Canada.

Study Selection

Following the search, all identified citations will be collated and uploaded to EndNote online (Clarivate Analytics, PA, USA), and duplicates will be removed.

Following a pilot test, titles and abstracts will be screened by two independent reviewers for assessment of the inclusion criteria for the review. Potentially relevant studies will be retrieved in full, and their citation details will be imported into the JBI System for the Unified Management, Assessment, and Review of Information (JBI SUMARI; JBI, Adelaide, Australia)[33].

The full text of the selected citations will be assessed in detail against the inclusion criteria by two independent reviewers. The reasons for excluding full-text studies that did not meet the inclusion criteria will be recorded and reported in the systematic review. Any

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disagreements between the reviewers at each stage of the study selection process will be resolved through discussion or by a third reviewer.

The search results, study selection, and inclusion process will be reported in full in the final systematic review and presented in preferred reporting items for systematic reviews and meta-analyses (PRISMA) flow diagram[34].

Assessment of Methodological Quality

Two independent reviewers will critically appraise eligible studies at the study level for methodological quality in the review using standardised critical appraisal instruments from JBI for experimental, quasi-experimental, and observational studies (see Online Supplemental Material Appendix 2 – Critical Appraisal Tools)[28]. Authors of papers will be contacted to request missing or additional data for clarification, where required.

Any disagreements that arise will be resolved through discussion or with a third reviewer.

Regardless of methodological quality, all studies will undergo data extraction and synthesis (where possible). A table accompanying a narrative will report the results of critical appraisal.

Data Extraction

Data will be extracted from studies included in the review by two independent reviewers using an adapted version of the JBI data extraction tool[27], including specific information related to the characteristics of the intervention (see Online Supplemental Material Appendix 3 –Extraction Tool). The data extract will also include specific details about the study author(s), country, characteristics of higher education institution, setting of the intervention, characteristics of participants (including year and field of study), study design, description of the intervention (including the type of educational method and duration of the intervention), outcomes of significance to the review question assessed (knowledge, skills, attitudes, and behaviours), follow-up time, main results, limitations, and additional data when required. As recommended by Redd et al. (2005)[35] the

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description and information on the reliability and validity of instruments used to evaluate the intervention will also be extracted.

Data Synthesis

Literature has suggested that it is challenging to synthesise data from educational interventions owing to heterogeneities in interventions and studies methodologies[36, 37]. Therefore, data will be initially assessed for similarity or extent of variation in outcome measures, measurement scales, and type of interventions before determining whether it is appropriate to enter into a meta-analysis. If it is considered suitable, data will be entered into a meta-analysis using JBI SUMARI.

Effect sizes will be expressed as either odds ratios (for dichotomous data) or weighted (or standardised) final post-intervention mean differences (for continuous data), and their 95% confidence intervals will be calculated for analysis. If meta-analysis can be conducted, heterogeneity will be assessed statistically using the standard χ^2 and I^2 tests. Statistical analyses will be performed using a random effects model[38] to allow generalisation.

To investigate potential sources of heterogeneity, subgroup analyses will be performed based on the year of the study, the field of nursing if applicable, intervention delivery (teaching method), the content of the intervention (patient safety topics), and the time frame of the intervention. Sensitivity analyses will be conducted to test decisions made regarding the effectiveness of interventions.

A funnel plot will be generated using RevMan 5.3 (Copenhagen: The Nordic Cochrane Centre, Cochrane) to assess publication bias if ten or more studies are included in the meta-analysis. Statistical tests for funnel plot asymmetry (Egger test) will be performed where appropriate.

If there is significant variation in the data and statistical pooling is not possible, the findings will be presented in narrative form, including tables and figures to enhance data presentation[38].

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Assessing Certainty of Findings

The GRADE approach for grading the certainty of evidence will be followed[39], and a summary of findings will be created using GRADEpro GDT V5 (McMaster University, ON, Canada).

The Summary of Findings will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence based on the risk of bias, directness, heterogeneity, precision, and risk of publication bias of the review results. The outcomes reported in the Summary of Findings will be knowledge, skills, attitudes, and behaviours of undergraduate nursing students related to patient safety.

PATIENT AND PUBLIC INVOLVEMENT

There will be no patient or public involvement in the conduct or dissemination of the results of this study.

ETHICS AND DISSEMINATION

This review does not require ethical approval because the primary population data will not be collected. This protocol complies with PRISMA guidelines. The findings will be disseminated at professional conference presentations and publications in pre-print and peer-reviewed open access journals.

COMPETING INTERESTS

The authors declare no conflict of interest, financial or otherwise.

FUNDING

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AUTHORS' CONTRIBUTIONS

HDR conceptualised the study with inputs from AG and AN. HDR designed and drafted the review protocol. AM, AV, AG, AN, EQ, and DL contributed to the protocol development. MS provided critical revision of the drafts for important intellectual content. All authors provided input to the drafts of the manuscript and approved the final version of this manuscript.

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REFERENCES

1. World Health Organization (WHO). Global Patient Safety Action Plan 2021–2030: Towards eliminating avoidable harm in health care. Geneva, Switzerland: World Health Organization 2021. Available https://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan [accessed Jan 2021].

2. World Health Organization (WHO). Ten facts on patient safety. 2019.Available https://www.who.int/news-room/facts-in-pictures/detail/patient-safety[accessed July 2021].

3. Wu AW, Busch IM. Patient safety: a new basic science for professional education. *GMS J Med Educ* 2019;36(2):Doc21.doi:10.3205/zma001229.

4.Bates DW, Singh H. Two Decades Since To Err Is Human: An Assessment Of Progress And Emerging Priorities In Patient Safety. *Health Aff* 2018;37(11):1736-43. doi:10.1377/hlthaff.2018.0738.

5. Kuriakose R, Aggarwal A, Sohi RK, et al. Patient safety in primary and outpatient health care. *J Family Med Prim Care* 2020;9:7-11.doi:10.4103/jfmpc.jfmpc_837_19.

6. National Academies of Sciences, Engineering, and Medicine- EAM. Crossing the global quality chasm: Improving health care worldwide. Washington, DC: The National Academies Press 2018.doi:10.17226/2515.

7. Slawomirski L, Auraaen A, Klazinga N. The economics of patient safety: Strengthening a value-based approach to reducing patient harm at national level. *OECD Health Working Papers* 2017;96.doi: 10.1787/5a9858cd-en.

8. Aiken LH, Sloane DM, Bruyneel L, et al. Nurse staffing and education and hospital mortality in nine European countries: a retrospective observational study. *Lancet* 2014;383(9931):1824-30. doi:10.1016/S0140-6736(13)62631-8 .

9. Oldland E, Botti M, Hutchinson AM, et al. A framework of nurses' responsibilities for quality healthcare — Exploration of content validity. *Collegian* 2020;27(2):150-63. doi:10.1016/j.colegn.2019.07.007[published Online First: 20 November 2019].

10. World Health Organization (WHO). Nursing and midwifery. 2020.https://www.who.int/news-room/fact-sheets/detail/nursing-and-midwifery[accessed May 2021].

11. Vaismoradi M, Tella S, Logan PA, et al. Nurses' Adherence to Patient Safety Principles: A Systematic Review. *Int J Environ Res Public Health* 2020;17(6):2028.doi: 10.3390/ijerph17062028.

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12. Mansour MJ, Al Shadafan SF, Abu- Sneineh FT, AlAmer MM. Integrating Patient Safety Education in the Undergraduate Nursing Curriculum: A discussion Paper. *Open Nurs J* 2018;12:125-32.doi:10.2174/1874434601812010125.

13. Dimitriadou M, Merkouris A, Charalambous A, et al. The knowledge about patient safety among undergraduate nurse students in Cyprus and Greece: a comparative study. *BMC Nurs* 2021;20(110).doi:10.1186/s12912-021-00610-6.

14. QSEN Institute. https://qsen.org/ (accessed July 2021).

15. Cronenwett L, Sherwood G, Barnsteine J, et al. Quality and safety education for nurses. *Nurs Outlook* 2007;55(3):122-31.doi:10.1016/j.outlook.2007.02.006.

16. World Health Organization (WHO). WHO Patient safety curriculum guide: Multi-professional edition. Geneva, Switzerland: World Health Organization 2011.https://apps.who.int/iris/bitstream/handle/10665/44641/9789241501958_eng.pdf (accessed Jul 2021).

17. Levett-Jones T, Dwyer T, Reid-Searl K, et al. Patient safety competency framework (PSCF) for nursing students. 2017.https://www.cqu.edu.au/__data/assets/pdf_file/0026/65780/PatientSafetyCompetencyFrameworkFINAL.pdf(accessed Jul 2021).

18. Murray M, Sundin D, Cope V. New graduate registered nurses' knowledge of patient safety and practice: A literature review. *J Clin Nurs* 2018;27:31-47.doi:10.1111/jocn.13785.

19. Garzin ACA, Melleiro MM. Safety in the training of health professionals. *Cienc Cuid Saude* 2019;18(4):e45780.doi:10.4025/cienccuidsaude.v18i4.45780.

20. Rebeschi LM. Perceived patient safety competence of baccalaureate nursing students: A descriptive comparative study. *SAGE Open Nurs* 2020;6:2377960820930134. doi:10.1177/2377960820930134

21. Tella S, Liukka M, Jamookeeah D, et al. What do nursing students learn about patient safety? An integrative literature review. *J Nurs Educ* 2014;53(1):7-13.doi:10.3928/01484834-20131209-04.

22. Usher k, Woods C, Conway J, et al. Patient safety contente and delivery in pre-registration nursing curricula: A national cross-sectional survey study. *Nurse Educ Today* 2018;66:82-89.doi:10.1016/j.nedt.2018.04.013.

23. Kirwan M, Riklikiene O, Gotlib J, et al. Regulation and current status of patient safety content in pre-registration nurse education in 27 countries: Findings from the rationing - missed nursing care (RANCARE) COST action project. *Nurse Educ Pract* 2019;37:132-140.doi:10.1016/j.nepr.2019.04.013.

For peer review only

24. VanGraafeiland B, Sloand E, Silbert-Flagg J, et al. Academic-clinical service partnerships are innovative strategies to advance patient safety competence and leadership in prelicensure nursing students. *Nurs Outlook* 2019;67(1):49-53.doi:10.1016/j.outlook.2018.08.003 .

25. Bianchi M, Bressan V, Cadorin L, et al. Patient safety competencies in undergraduate nursing students: a rapid evidence assessment. *J Adv Nurs* 2016;72(12):2966-2979.doi:10.1111/jan.13033.

26. Lee SE, Morse BL, Kim NW. Patient safety educational interventions: A systematic review with recommendations for nurse educators. *Nurs Open* 2021;00:1–13.doi:10.1002/nop2.955.

27. Tufanaru C, Munn Z, Aromataris E, et al. Chapter 3: Systematic reviews of effectiveness. In: Aromataris E, Munn Z, eds. JBI Manual for Evidence Synthesis: JBI 2020.https://synthesismanual.jbi.global/ (accessed Jun 2021).

28. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71.doi:10.1136/bmj.n71.

29. McKenzie JE, Brennan SE, Ryan RE, et al. Chapter 3: Defining the criteria for including studies and how they will be grouped for the synthesis. In: Higgins JPT, Thomas J, Chandler J, et al, eds. Cochrane Handbook for Systematic Reviews of Interventions version 6.2 (updated February 2021): Cochrane 2021.www.training.cochrane.org/handbook (accessed Jul 2021).

30. Stern C, Jordan Z, McArthur A. Developing the review question and inclusion criteria. *Am J Nurs* 2014;114(4):53-6.doi:10.1097/01.NAJ.0000445689.67800.86.

31. Yu A. Fontana G. Darzi A. Evaluation and training interventions for patient safety: A report by the centre for health policy at Imperial College London, an academic partner to Health Education England and the commission on education and training for patient safety 2016. https://www.hee.nhs.uk/sites/default/files/documents/Evaluation%20of%20Education%20and%20Training%20Interventions%20for%20Patient%20Safety%20-%20Imperial%20College%20London.pdf(accessed Jul 2021).

32. Aromataris E, Munn Z (Editors). JBI Manual for Evidence Synthesis. JBI, 2020.https://synthesismanual.jbi.global (accessed Jun 2021).

33. Munn Z, Aromataris E, Tufanaru C, et al. The development of software to support multiple systematic review types: the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). *Int J Evid Based Healthc* 2019;17(1):36-43.doi:10.1097/xeb.0000000000000152.

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34. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71.doi:10.1136/bmj.n71.

35. Reed D, Price EG, Windish DM, et al. Challenges in systematic reviews of educational intervention studies. *Ann Intern Med* 2005;142(12Pt2):1080-9.doi:10.7326/0003-4819-142-12_Part_2-200506211-00008.

36. Slavin RE. Perspectives on evidence-based research in education—What works? Issues in synthesizing educational program evaluations. *Educ Res* 2008;37(1):5-14.doi:10.3102/0013189X08314117 [published Online First: 21 June 2005].

37. Tufanaru C, Munn Z, Stephenson M, et al. Fixed or random-effects meta-analysis? Common methodological issues in systematic reviews of effectiveness. *Int J Evid Based Healthc* 2015;13(3):196–207.doi:10.1097/xeb.0000000000000065.

38. Campbell M, McKenzie JE, Sowden A, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. BMJ. 2020; 368: l6890.doi:10.1136/bmj.l6890.

39. Guyatt G, Oxman AD, Akl EA, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol* 2011;64(4):383–94.doi:10.1016/j.jclinepi.2010.04.026.

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Appendix 1

Search Strategies

Database	#	Search Strategy
1) Medline (PubMed)		(((student, nursing[MeSH Terms]) OR (((((("nursing students"[Text Word]) OR ("pupil nurses"[Text Word])) OR ("undergraduate nurses"[Text Word])) OR ("baccalaureate nurses"[Text Word])) OR ("student nurses"[Text Word])) OR ("undergraduate student nurses"[Text Word])) OR (pre licensure nurses[Text Word]))) AND (("education"[MeSH Terms]) OR (((((("education"[Text Word]) OR ("educational activities"[Text Word])) OR (training program*[Text Word])) OR ("learning"[Text Word])) OR ("teaching"[Text Word])) OR (teaching strateg*[Text Word])) OR ("teaching methods"[Text Word]))))) AND (("patient safety"[MeSH Terms]) OR ((("patient safety"[Text Word]) OR ("patient safety competency"[Text Word])) OR ("error"[Text Word])) OR ("incident"[Text Word]))) Filters: English, Portuguese, Spanish, from 2011/7 - 2022/1
2) CINAHL and Education Research Complete (via EBSCOhost)	**#1** **#2** **#3** **#4**	((MM "Students, Nursing") OR (MH "Students, Nursing, Baccalaureate")) OR TX "nursing students" OR TX "pupil nurses" OR TX "undergraduate nurses" OR TX "baccalaureate nurses" OR TX "student nurses" OR TX "undergraduate student nurses" ((MM "Education") OR (MH "Education, Nursing, Graduate")) OR TX "education" OR TX "educational activities" OR TX training program* OR TX "learning" OR TX "teaching" OR TX teaching strateg* ((MM "Patient Safety") OR (MH "Health Care Errors")) OR TX "patient safety competency" OR TX error* OR TX "incident" #1 AND #2 AND #3 Limiters - Published Date: 20110701-20220131 Narrow by Language: - Spanish, Portuguese, English
3) Scopus (Elsevier)		(TITLE-ABS-KEY({nursing students} OR {nursing pupils} OR {undergraduate nurses} OR {baccalaureate nurses} OR {student nurses} OR {undergraduate student nurses} OR {pre-licensure nurses}) AND TITLE-ABS-KEY({patient safety}) AND ALL({education} OR {learning} OR {teaching})) AND PUBYEAR > 2010 AND (LIMIT-TO (LANGUAGE,"English") OR LIMIT-TO (LANGUAGE,"Portuguese") OR LIMIT-TO (LANGUAGE,"Spanish"))
4) Cochrane (Central)	**#1** **#2** **#3** **#4** **#5** **#6** **#7** **#8** **#9** **#10** **#11** **#12**	MeSH descriptor: [Students, Nursing] explode all trees (nursing students) OR (pupil nurses) OR (undergraduate nurses) OR (baccalaureate nurses) OR (student nurses) OR (undergraduate student nurses) OR (pre-licensure nurses) #1 OR #2 MeSH descriptor: [Education] explode all trees MeSH descriptor: [Education, Nursing] explode all trees #4 OR #5 (education) OR (educational activities) OR (training program) OR (learning) OR (teaching) OR (teaching strategy) OR (teaching methods) #6 OR #7 MeSH descriptor: [Patient Safety] explode all trees (patient safety) OR (patient safety competency) OR (error) OR (incident) #9 OR #10 #3 AND #8 AND #11 with Publication Year from 2011 to 2022
5) Lilacs		(mh:("Seguridad del Paciente")) AND (estudante* OR student OR estudiante) AND (educacao OR ensino OR aprendizagem) AND (enfermagem OR nursing OR enfermeria) AND (db:("LILACS"))

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6) Medes		((("estudiantes de enfermería"[todos]) AND "enseñanza"[todos]) AND "seguridad del paciente"[todos]) https://medes.com/Public/Home.aspx
7) Clinical Trials.gov		condition or disease: patient safety other terms: nursing students intervention treatment: education https://www.clinicaltrials.gov/
8) Google Scholar		"patient safety" AND "education" and "nursing undergraduates" OR "student nurses" OR "nursing students" AND filetype: PDF OR RTF OR DOC OR TXT OR html
9) Dart-Europe		nursing students AND education AND patient safety https://www.dart-europe.org/basic-search.php
10) Proquest Dissertations		ft("nursing students" OR "undergraduate nursing student" OR "pupil nurse" OR "student nurse") AND ft("education" OR "training" OR "teaching" OR teaching program* OR "teaching methods") AND ft("patient safety" OR "patient safety competency") Additional limits - Date: From July 2011 to January 2022 Language: English, Portuguese, Spanish
11) CAPES thesis and dissertations		"estudantes de enfermagem" AND "educação" OR "ensino" OR "aprendizagem" AND "segurança do paciente" https://catalogodeteses.capes.gov.br/catalogo-teses/
12) The Virginia Henderson Global e-Repository		"nursing students" OR "undergraduate nursing student" OR "pupil nurse" OR "student nurse" AND "education" OR "training" OR "teaching" OR teaching program* OR "teaching methods" AND "patient safety" OR "patient safety competency" https://sigma.nursingrepository.org/discover#showfilters
13) Mednar		nursing students OR undergraduate nursing student OR pupil nurse OR student nurse AND education OR teaching OR teaching methods AND patient safety https://mednar.com/mednar/desktop/en/search.html?pane=advanced
14) Thesis Canada		nursing students OR student nurse AND education OR teaching AND patient safety https://www.bac-lac.gc.ca/eng/services/theses/Pages/search.aspx

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Appendix 2

Critical Appraisal Tools

JBI CRITICAL APPRAISAL CHECKLIST FOR RANDOMIZED CONTROLLED TRIALS

Reviewer_____

Date_____

Author_____Year_____ Record Number_____

		Yes	No	Unclear	NA
1.	Was true randomization used for assignment of participants to treatment groups?	☐	☐	☐	☐
2.	Was allocation to treatment groups concealed?	☐	☐	☐	☐
3.	Were treatment groups similar at the baseline?	☐	☐	☐	☐
4.	Were participants blind to treatment assignment?	☐	☐	☐	☐
5.	Were those delivering treatment blind to treatment assignment?	☐	☐	☐	☐
6.	Were outcomes assessors blind to treatment assignment?	☐	☐	☐	☐
7.	Were treatment groups treated identically other than the intervention of interest?	☐	☐	☐	☐
8.	Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	☐	☐	☐	☐
9.	Were participants analyzed in the groups to which they were randomized?	☐	☐	☐	☐
10.	Were outcomes measured in the same way for treatment groups?	☐	☐	☐	☐
11.	Were outcomes measured in a reliable way?	☐	☐	☐	☐
12.	Was appropriate statistical analysis used?	☐	☐	☐	☐
13.	Was the trial design appropriate, and any deviations from the standard RCT design (individual randomization, parallel groups) accounted for in the conduct and analysis of the trial?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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JBI CRITICAL APPRAISAL CHECKLIST FOR QUASI-EXPERIMENTAL STUDIES

Reviewer_____

Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	□	□	□	□
2. Were the participants included in any comparisons similar?	□	□	□	□
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	□	□	□	□
4. Was there a control group?	□	□	□	□
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	□	□	□	□
6. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	□	□	□	□
7. Were the outcomes of participants included in any comparisons measured in the same way?	□	□	□	□
8. Were outcomes measured in a reliable way?	□	□	□	□
9. Was appropriate statistical analysis used?	□	□	□	□

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

For peer review only

JBI CRITICAL APPRAISAL CHECKLIST FOR CASE SERIES

Reviewer_____
Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
• Were there clear criteria for inclusion in the case series?	□	□	□	□
• Was the condition measured in a standard, reliable way for all participants included in the case series?	□	□	□	□
• Were valid methods used for identification of the condition for all participants included in the case series?	□	□	□	□
• Did the case series have consecutive inclusion of participants?	□	□	□	□
• Did the case series have complete inclusion of participants?	□	□	□	□
• Was there clear reporting of the demographics of the participants in the study?	□	□	□	□
• Was there clear reporting of clinical information of the participants?	□	□	□	□
• Were the outcomes or follow up results of cases clearly reported?	□	□	□	□
• Was there clear reporting of the presenting site(s)/clinic(s) demographic information?	□	□	□	□
• Was statistical analysis appropriate?	□	□	□	□

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

JBI CRITICAL APPRAISAL CHECKLIST FOR COHORT STUDIES

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Reviewer_____

Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the two groups similar and recruited from the same population?	☐	☐	☐	☐
2. Were the exposures measured similarly to assign people to both exposed and unexposed groups?	☐	☐	☐	☐
3. Was the exposure measured in a valid and reliable way?	☐	☐	☐	☐
4. Were confounding factors identified?	☐	☐	☐	☐
5. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	☐	☐	☐	☐
7. Were the outcomes measured in a valid and reliable way?	☐	☐	☐	☐
8. Was the follow up time reported and sufficient to be long enough for outcomes to occur?	☐	☐	☐	☐
9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	☐	☐	☐	☐
10. Were strategies to address incomplete follow up utilized?	☐	☐	☐	☐
11. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

For peer review only

JBI CRITICAL APPRAISAL CHECKLIST FOR CASE CONTROL STUDIES

Reviewer_____
Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the groups comparable other than the presence of disease in cases or the absence of disease in controls?	☐	☐	☐	☐
2. Were cases and controls matched appropriately?	☐	☐	☐	☐
3. Were the same criteria used for identification of cases and controls?	☐	☐	☐	☐
4. Was exposure measured in a standard, valid and reliable way?	☐	☐	☐	☐
5. Was exposure measured in the same way for cases and controls?	☐	☐	☐	☐
6. Were confounding factors identified?	☐	☐	☐	☐
7. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
8. Were outcomes assessed in a standard, valid and reliable way for cases and controls?	☐	☐	☐	☐
9. Was the exposure period of interest long enough to be meaningful?	☐	☐	☐	☐
10. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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JBI CRITICAL APPRAISAL CHECKLIST FOR ANALYTICAL CROSS SECTIONAL STUDIES

Reviewer_____
Date_____

Author_____Year_____ Record Number_____

	Yes	No	Unclear	Not applicable
1. Were the criteria for inclusion in the sample clearly defined?	☐	☐	☐	☐
2. Were the study subjects and the setting described in detail?	☐	☐	☐	☐
3. Was the exposure measured in a valid and reliable way?	☐	☐	☐	☐
4. Were objective, standard criteria used for measurement of the condition?	☐	☐	☐	☐
5. Were confounding factors identified?	☐	☐	☐	☐
6. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
7. Were the outcomes measured in a valid and reliable way?	☐	☐	☐	☐
8. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____
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_____

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Appendix 3
Extraction Tool

Extraction Tool
Study refence (Author/year) and Study ID
Country: **Type of Higher Education Institution:** Public ( ) Private ( ) Other ( )
Setting of the intervention delivery: University ( ) Clinical Setting ( ) Both Combined ( )
Description of the intervention: Type of educational method (s): Duration of the intervention:
Group description: Intervention: Control:
Sample Size: Intervention: Control:
Outcomes of interest assessed: Knowledge ( ) Skills ( ) Attitudes ( ) Behaviours ( )
Name and description of instrument for outcome assessment: Reliability: Validity:
Follow-up time:
Main Results:
Limitations:

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page/Line Number Marked Copy
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1 (lines 1-3)
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 2 (line 55)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1 (lines 5-23)
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 14 (lines 321-326)
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Page 14 (lines 318-319)
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 5 (lines 116-121) Page 6 (137-145)
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 7 (lines 157-159)
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Pages 6-7 (lines 166-195 Page 10 (lines 209-212)
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 10 (lines 212-223)
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it	Supplemental material

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		could be repeated	
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 10 (lines 226-232)
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Page 10 (lines 233-237)
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 10 (lines 254-257)
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Pages 11-12 (lines 257-265)
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 8 (lines 185-188)
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 11 (lines 243-251)
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	Page 12 (lines 270-273)
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Page 12 (lines 274-279)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	Page 12 (lines 280-284)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Page 12 (lines 289-291)
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	Page 12 (lines 285-288)
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Page 13 (lines 293-302)

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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